

Green ammonia at Sines — 500 t/day synthesis loop

PEQ 2026–27 project brief, revised edition. Companion document: *Thermodynamic basis* (thermoBasis.pdf).

Why this edition exists

The first edition of this brief asked you to appraise the plant with hydrogen bought at 4.00 EUR/kg and ammonia sold at 0.60 EUR/kg. Run that pair through the case and the answer is a net present value of **−680 million EUR**: the plant loses more than its entire fixed capital investment every year, and no decision you can take inside the battery limits changes that: the Converter is the most expensive item in the plant and its whole total-module cost is **8.4 million EUR**, against a hole of 680.

That is not a defect of the flowsheet. It is arithmetic, and it is worth seeing in one line. The case consumes **181.1 kg of hydrogen per tonne of ammonia** (measured; the stoichiometric minimum is 177.6, the extra 2.0% is the loop purge). At 4.00 EUR/kg that is **725 EUR of hydrogen inside each tonne of ammonia**, sold for 600. You lose 0.12 EUR on every kilogram before touching capital, utilities, labour or catalyst.

The lesson is real and it is the most important thing this project can teach you — **but it is a lesson about a price pair, not about a design**, and a brief that hides it behind a fixed price list teaches the wrong reflex. This edition therefore makes the hydrogen price an *object of the study* instead of a given, and asks you to defend the number you choose.

1 What you are designing — and what you are not

Battery limits. The electrolyser park and the air separation unit are **outside**. Hydrogen and nitrogen arrive at the fence as purchased raw materials. Everything this case sizes, costs and appraises is the **synthesis loop plus its compression and purification**.

This matters more than it looks. The fixed capital investment the case computes is **29.9 million EUR**; the raw-material bill at the first edition's prices is **129.3 million EUR per year**. The capital you are designing is worth about three months of the hydrogen you are buying. **The investment decision for a green ammonia plant lives in the electricity-to-hydrogen step, which is outside your fence** — and one of the deliverables below is to show that quantitatively rather than to be told it.

The four sectors

Sector	What it does
PURIFICATION	Compresses and cools the nitrogen, mixes it with hydrogen, burns out the oxygen the industrial-grade nitrogen brings (Deoxo), chills and knocks out the water that reaction makes, dries the gas over a 4A sieve, and compresses the make-up to loop pressure.
LOOP	Recycle compressor, mixer, feed–effluent exchanger, adiabatic Gibbs converter, water cooler, cold separator, purge splitter.

PRODUCT	Lets the raw liquid ammonia down to 20 bar and flashes off the dissolved gases.
RECOVERY	Lets the purge down, warms it and scrubs the ammonia out of it with water.

2 Design basis (the declared inputs)

These are the numbers the case *declares*. They are yours to change — that is the point of the project — but change them deliberately and say so.

Quantity	Value	Where it is declared
Ammonia production (target)	500 t/day	the design target
Nitrogen quality	99.50 % N ₂ , 0.40 % O ₂ , 0.10 % Ar	O/MAIN/N2Feed
Hydrogen delivery pressure	30 bar	O/MAIN/H2Feed
Loop pressure	200 bar	LOOP/Converter, LOOP/Separator
Converter	adiabatic Gibbs equilibrium	LOOP/Converter
Separator temperature	250 K	LOOP/Separator
Purge fraction	0.2 % of the recycle	LOOP/PurgeSplit
Deoxo conversion	1.0 on O ₂ (limiting)	PURIFICATION/Deoxo
Dryer	4A sieve, 1500 kg/bed, 12 h cycle	PURIFICATION/Dryer
Shaft power installed	908 / 6700 / 1400 kW	the three compressors
Operating hours	7998 h/yr (stream factor 0.913)	system/postDict
Discount rate, project life	8 %, 20 years	system/postDict

The nitrogen is the interesting declaration. Analytical-grade nitrogen at 99.999 % is what you buy in a cylinder to feed a chromatograph. This plant consumes 17 t/h of it: at that rate you take what the air separation column gives you, which is 99.5 to 99.9 %, with oxygen as the dominant impurity. Three units in PURIFICATION exist only because of that one line, and the argon in it builds up in the loop to **1.38 mol %** at the converter — which is what sets the purge, which is what sets the hydrogen you throw away.

3 What the plant does (measured)

Every figure below is read off a Choupo run of this case. Reproduce them with `runCase -f` in the case directory.

Quantity	Symbol	Value
Ammonia produced	NH3_t_day	499.85 t/day
Hydrogen consumed	H2_specific_kg_t	181.15 kg/t NH ₃
Total shaft power	W_total_kW	9008 kW
Specific compression power	W_specific_kWh_t	432.5 kWh/t NH ₃
Ammonia at the converter outlet	y_NH3_converter	11.01 mol %
Converter outlet temperature	T_converter_K	839.0 K
Argon at the converter	y_Ar_converter	1.38 mol %
Purge	purge_kmol_h	23.05 kmol/h
Oxygen removed in the deoxo	O2_removed_kmol_h	2.594 kmol/h
Water knocked out	H2O_condensed_kg_h	81.05 kg/h
Chilling duty	Q_chilling_kW	741.6 kW

Quantity	Symbol	Value
Dryer capacity used	dryer_utilisation	52.7 %

The plant closes its energy balance to **−12.70 kW**, which is **0.057 %** of the **22.4 MW** it exchanges. Choupo publishes both the residual and the scale it is judged against; do not take either on trust, read them in the run’s own **energyBalance** report.

Those two numbers used to read 728.9 kW and 0.074 % of 983.9 MW, and the second pair was the more misleading of the two. A chilled-water supply stream carried a **category** tag that its return did not, so 60 kg/s of water’s formation-datum enthalpy was booked as a 960 MW “duty” — which both inflated the scale forty-fold and double-counted 741.6 kW of chiller heat into the plant’s boundary term. **That double count WAS the residual.** Found by reading the panel and asking whether it looked right; fixed 2026-09-09.

4 The economics — and the price you have to defend

4.1 Two ways to get hydrogen, and they are not the same question

Scenario M (merchant). You buy hydrogen over the fence at a quoted price p_{H_2} [EUR/kg]. This is the case as it ships, and it is the right model for a plant that sits next to somebody else’s electrolyser.

Scenario I (integrated). The project owns the electrolyzers. There is then no market price — there is an electricity price and a conversion efficiency:

$$p_{\text{H}_2} = \frac{e p_e}{1000} + c_{\text{el}}, \quad (1)$$

where e is the specific consumption of the electrolyser [kWh per kg H₂], p_e the delivered electricity price [EUR/MWh] and c_{el} the share of the electrolyser’s own capital and maintenance carried by each kilogram [EUR/kg]. **Both e and c_{el} are yours to justify**, with a source; they are not Choupo data and this brief does not supply them.

4.2 The break-even, measured

Running the case across the price grid gives a strictly linear response (revenue is linear in the ammonia price, the raw-material bill linear in the hydrogen price), so the break-even is exact and can be quoted:

Ammonia price	Break-even p_{H_2}	Hydrogen bill at break-even
0.60 EUR/kg	1.64 EUR/kg	297 EUR/t NH₃ (49.5 % of revenue)
0.90 EUR/kg	2.99 EUR/kg	542 EUR/t NH₃ (60.2 % of revenue)

Break-even here means NPV = 0 at 8 % over 20 years, with this case’s own capital, labour and utility model. Verified by running the case at those two prices, not interpolated.

4.3 What that means in electricity

Substituting the break-even hydrogen price into equation (1) with $c_{\text{el}} = 0$ — that is, giving the electrolyser away for free — puts a hard ceiling on the electricity price the project can pay:

	$e = 50 \text{ kWh/kg}$	$e = 55 \text{ kWh/kg}$
Ceiling on p_e at NH_3 0.60 EUR/kg	32.8 EUR/MWh	29.8 EUR/MWh
Ceiling on p_e at NH_3 0.90 EUR/kg	59.8 EUR/MWh	54.4 EUR/MWh
Electrolysis power per tonne NH_3	9057 kWh/t	9963 kWh/t
<i>versus</i> compression, measured	432.5 kWh/t	

The two values of e in that table are round numbers chosen to bracket a plausible range. **They are not Choupo data and this brief does not cite a source for them** — supplying one is part of your work. Everything marked **bold** elsewhere in this document *is* a Choupo output and can be reproduced by running the case.

The last two rows are the whole argument. **The electrolysis consumes about twenty times the power of everything you are designing.** A student who spends the semester optimising the recycle ratio is polishing the third digit of a 4 % term.

5 Deliverables

1. **Reproduce the base case** and present the stream table, the per-unit and plant energy balances, and the mass closure. State the residuals; do not round them away.
2. **The design study.** Change the design decisions that are yours — loop pressure, separator temperature, purge fraction, exchanger areas, the design pressures and residence times in `system/postDict` — and report their effect on the fixed capital investment and on the specific power. At least one of your changes must make things *worse*, and you must explain why.
3. **The hydrogen price study.** Report NPV against p_{H_2} over the range 1.0 to 4.0 EUR/kg, at both ammonia prices, and mark the break-even. Reproduce the two break-even values in this brief or explain your disagreement.
4. **Scenario I.** Choose e and c_{el} , cite your sources, and state the delivered electricity price at which this plant breaks even. Then say plainly whether that price is reachable at Sines and on what evidence.
5. **The honest limitations section.** Read the run's own caveat block — Choupo prints every extrapolation, every unverified record and every assumed default at the end of the run — and turn it into a section of your report. Two of them are named in §6 below to start you off; the run lists **32**.

6 What this case does not model

Stated here so you do not have to discover it in a viva.

- **No credit is taken for anything the plant gives away.** The loop rejects **18.3 MW** at the water cooler and moves **42.1 MW** through the feed–effluent exchanger; a real plant exports part of that as steam. The oxygen by-product of electrolysis is outside the fence and is not sold. Both omissions push the appraisal *down*.
- **The utility bill is a declared lower bound.** The utility cost comes out at **13.53 million EUR/yr**, and **four** heat duties in this flowsheet reach no catalogue utility at all, so they are *not* in it. Every figure derived from it — COM_d , NPV, IRR — inherits that bound, and the run says so by name.

- **No regulatory or contractual framework** — RFNBO, CBAM, contracts for difference, capital grants. None of it is thermodynamics and none of it belongs in `constant/economics`. If your appraisal depends on it, say so explicitly.
- **The vessel sizer computes gas volumetric flow as NRT/P** , ideal gas, and states that in its own basis line. At 200 bar in the synthesis loop that is the one place in the sizing chain where the stated basis and the case's thermodynamic package genuinely disagree. Quantifying the error is a legitimate — and welcome — finding for your report.
- **SS316 is rated 100 bar in the material record** and the Converter and Separator are designed at 240 bar. Choupo warns per vessel and prices them anyway. Decide what to do about it and defend the decision.
- **Two boundary streams carry a phase label the energy report's own equilibrium check disputes** — `PURIFICATION.DryishGas` ($g(V=1) = -0.00138$, i.e. marginally below its dew point, which is what a flash's own vapour outlet is by construction) and `WashWater`. The report says so by name rather than absorbing it. Read the message; do not assume it names a defect in *your* case.
- **Construction is modelled as one year** even though `constructionPeriod 2` is declared; the whole investment is placed at $t = 0$. Announced by the run, not silent.

7 Running it

```
source /path/to/Choupo/etc/bashrc
cd tutorials/plant/greenAmmoniaIndustrialN2
runCase -f .           # solve, size, cost, appraise
choupoSolve --manifest . # where each stream's state file belongs
```

Outputs to read: `converged/` (the solved state, one file per stream), `design/<SECTOR>/<unit>/<item>` (one specification sheet per physical item), `reports/` (the balances, the utility allocation, the costing tables). The sizing, costing and economics chain is driven by `system/postDict` and priced by `constant/economics`; in both, a unit or a stream must be named by its **engine** identity — `LOOP.Converter`, not `converter`; `ProductNH3`, not `productNH3`.